

# Lesson 2: Energy per coulomb

Electric Circuits · Year 10 · 50 minutes · AQA 6.2.1.3, 6.2.4.2

## Why this lesson is here

The same charge can give a lamp more energy when the supply p.d. is higher. Voltage describes energy per coulomb, not the speed of the charges.

## Learning objective

Explain p.d. as energy transferred per coulomb and calculate  $E = QV$ .

## Core teaching point

Potential difference is energy transferred per unit charge:  $1\text{ V} = 1\text{ J/C}$ . A voltmeter is connected in parallel across the component.

## Equipment and preparation

Cells, two resistors or lamps, ammeter, voltmeter; PhET lab; virtual-lab worksheet pp. 7-11.

## Lesson sequence

| Time  | Phase         | Teacher move and evidence                                    |
|-------|---------------|--------------------------------------------------------------|
| 0-5   | Retrieval     | Compare charge, current and energy units.                    |
| 5-15  | Model         | Teach p.d. as J/C and correct meter placement.               |
| 15-32 | Virtual Lab 2 | Measure source and component p.d.s; check the sum.           |
| 32-43 | Practice      | Tasks 1, 2 and 4, then explain a voltmeter reading in words. |
| 43-50 | Check         | Mark four-mark question and identify unit errors.            |

## Workbook task guidance

Task 1: Draw a voltmeter across a lamp and an ammeter in series with it. Explain each position.

Expected: Ammeter in series, voltmeter parallel across lamp.

Task 2: Predict how the p.d. across a single lamp changes when the battery changes from 3 V to 6 V.

Expected: The p.d. across the only lamp approximately doubles, provided the supply is ideal.

Task 3: Virtual Lab 2: complete the p.d. and energy investigation on existing worksheet pp. 7-11. Record p.d. across the battery and two series components; test whether the drops add.

Expected: Battery p.d. approximately equals the sum of series component p.d.s; readings may differ slightly due to rounding/internal resistance.

Task 4: Calculate energy transferred when 25 C passes through a component with 4.0 V across it.

Expected:  $E = QV = 25 \times 4.0 = 100\text{ J}$ .

## Worked example

A lamp transfers 36 J as 6.0 C passes.  $V = E/Q = 36/6.0 = 6.0\text{ V}$ .

## AQA-style checkpoint

[4 marks] A 9.0 V battery moves 20 C of charge through a circuit. Calculate the energy transferred by the battery. Explain what a voltmeter reading of 3.0 V across one resistor means.

Mark guidance:  $E = QV$  (1) =  $20 \times 9.0 = 180\text{ J}$  (1). 3.0 V means 3.0 J transferred per coulomb (2).

### **Anticipated misconception**

Voltage is not 'used up': p.d. measures energy transfer between two points.

### **Assessment and next step**

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.